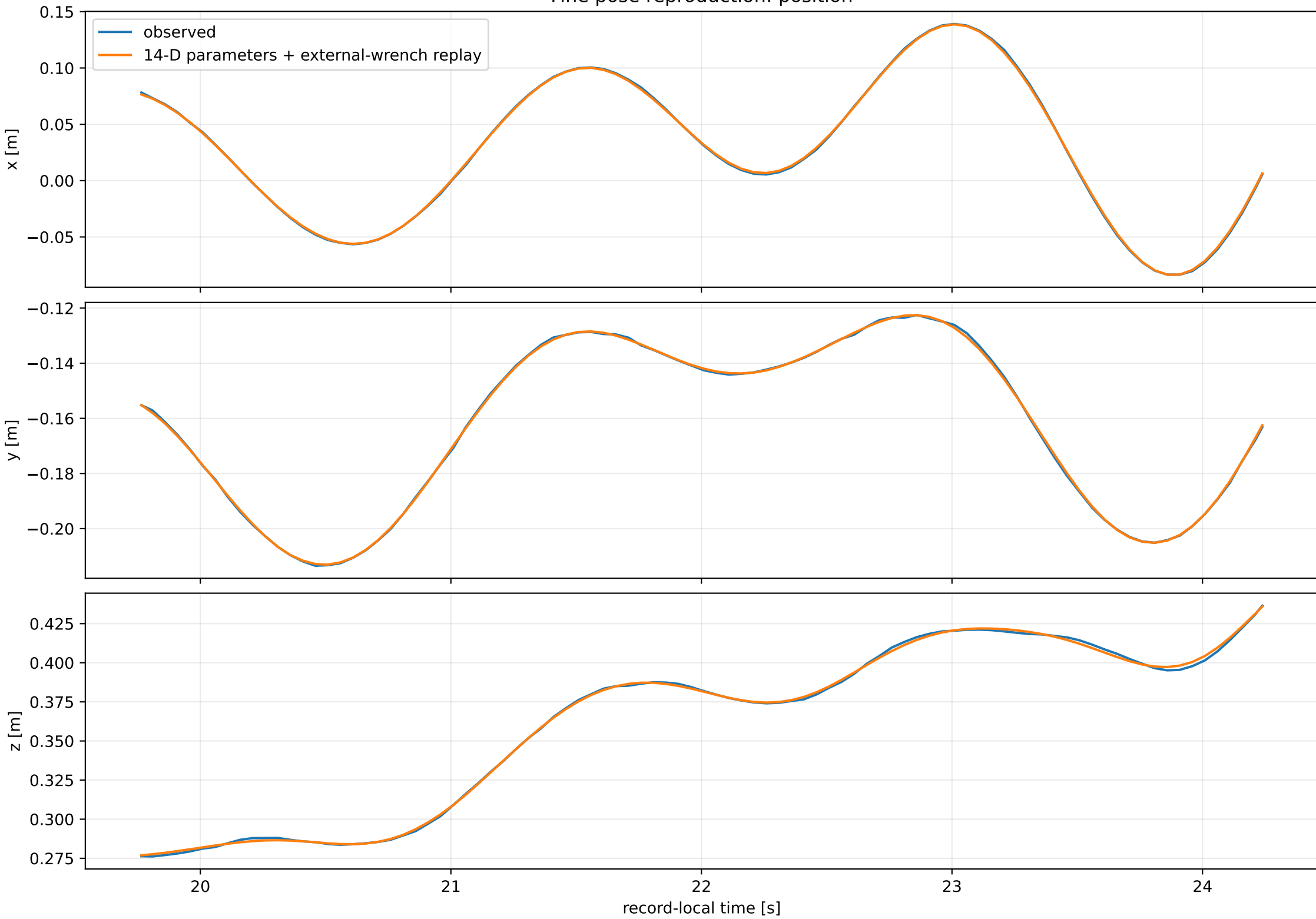
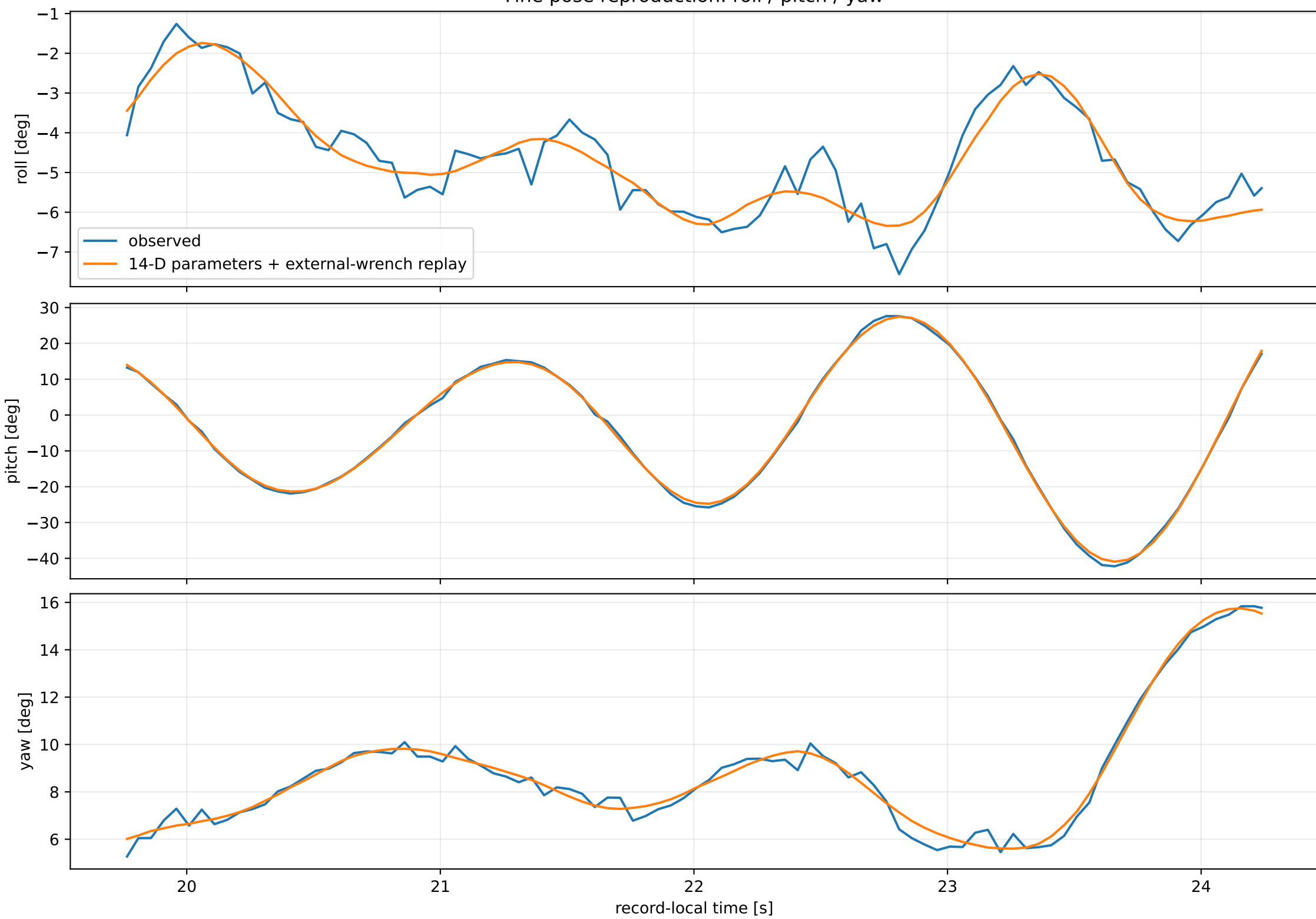


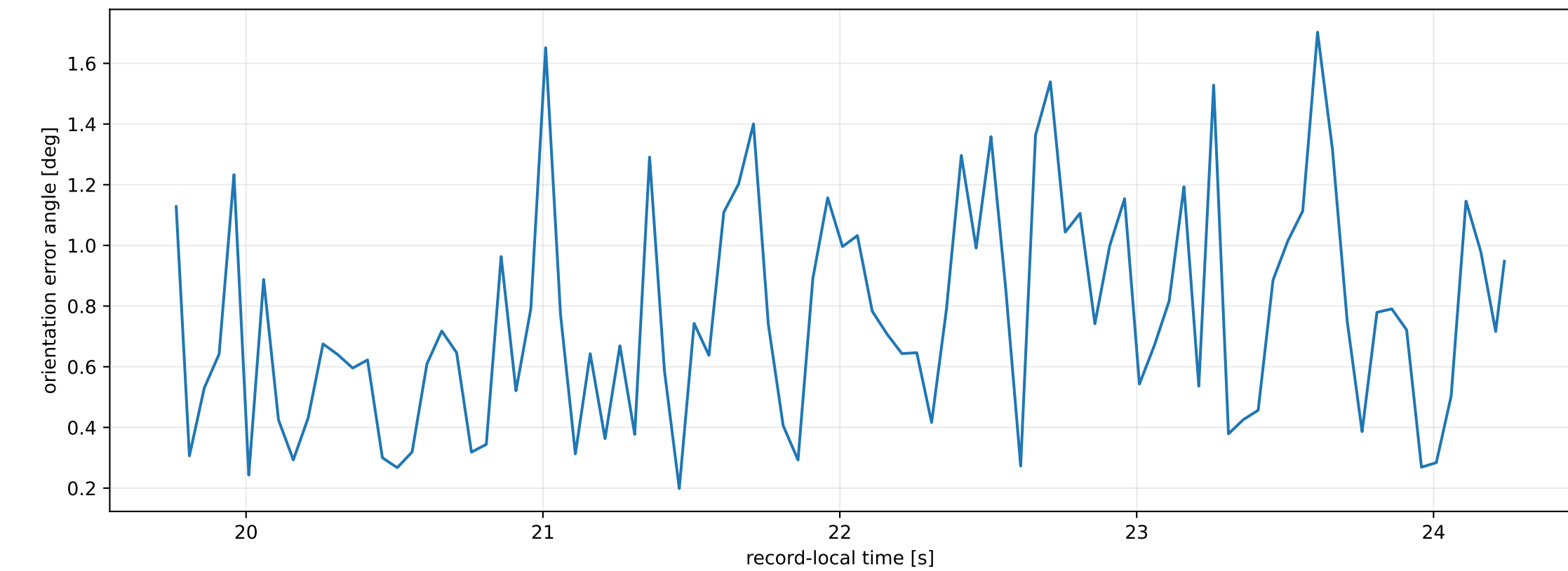
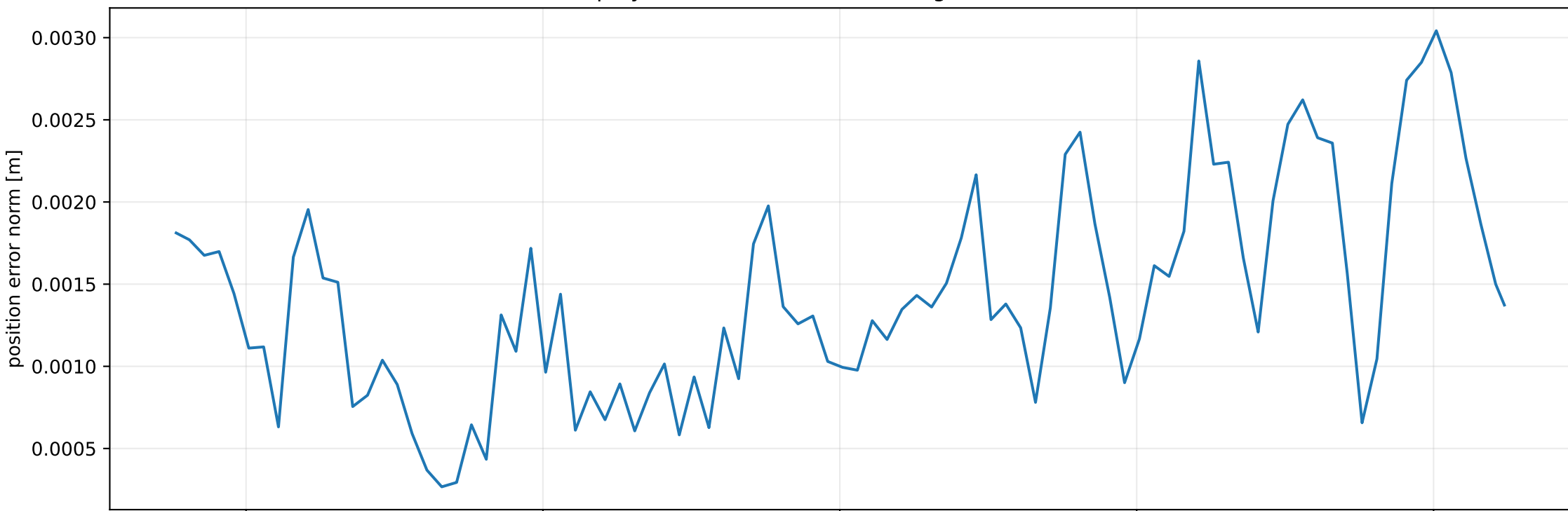
Fine pose reproduction: position



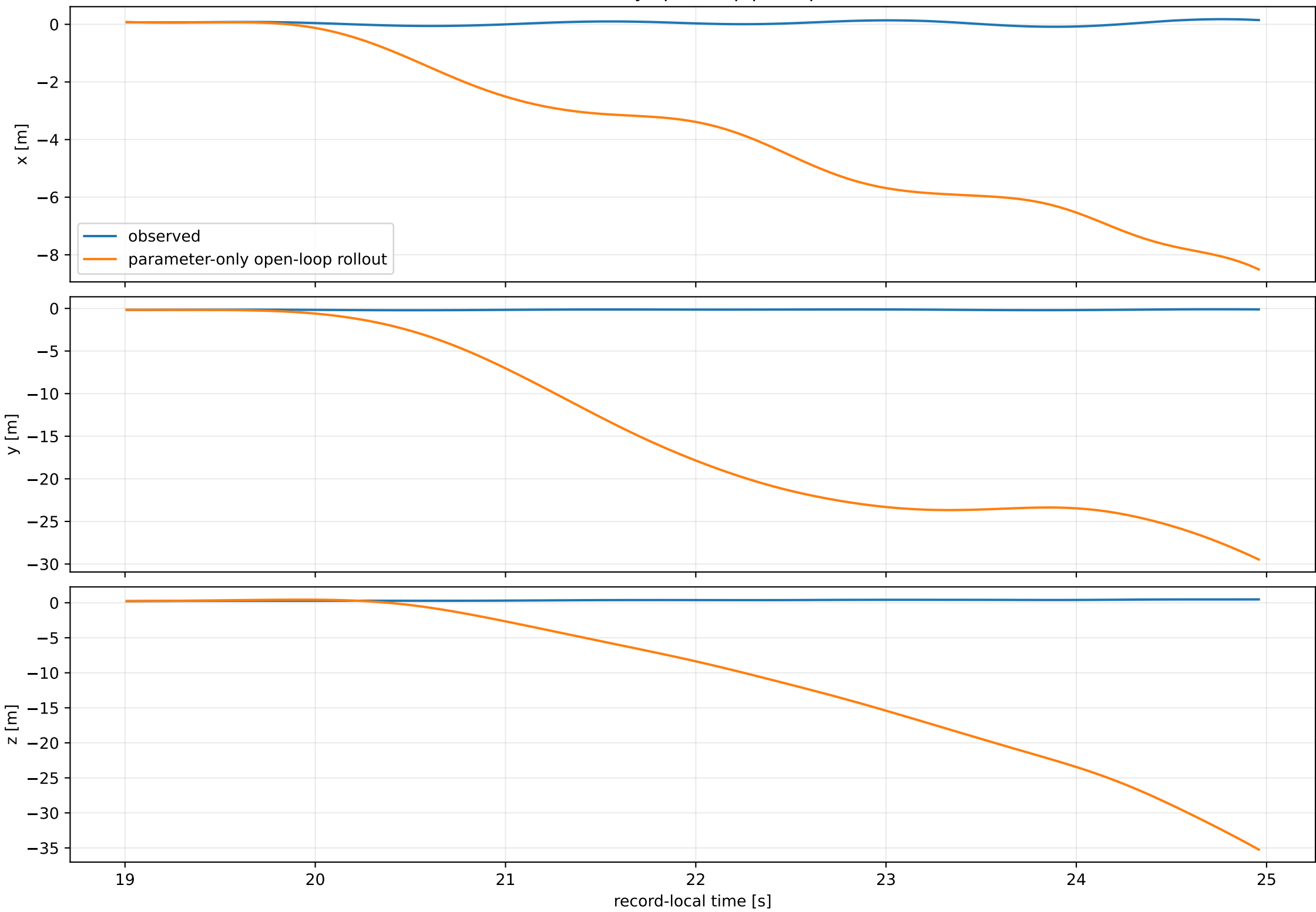
Fine pose reproduction: roll / pitch / yaw



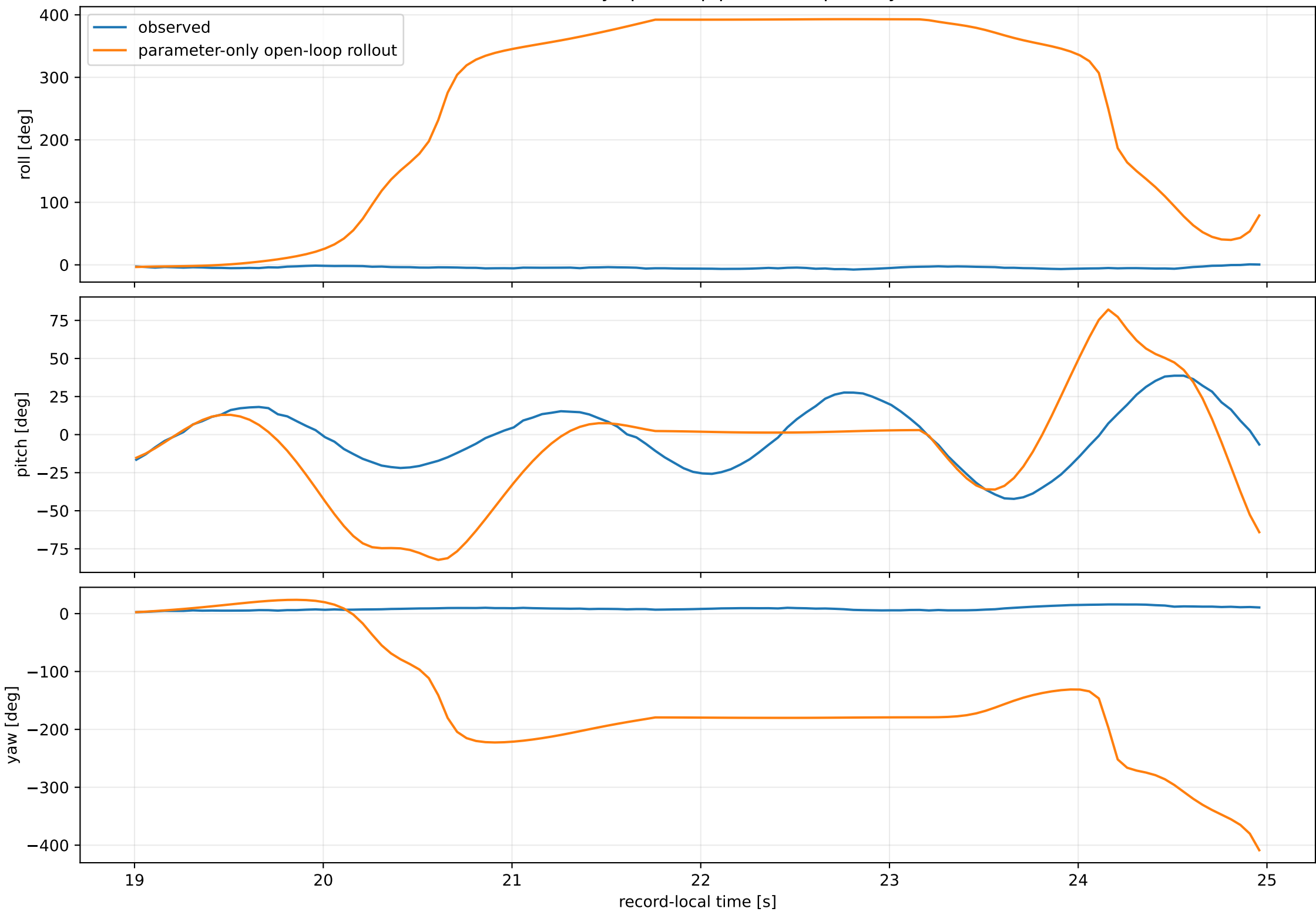
Fine replay error after the 14-D change — lower is better



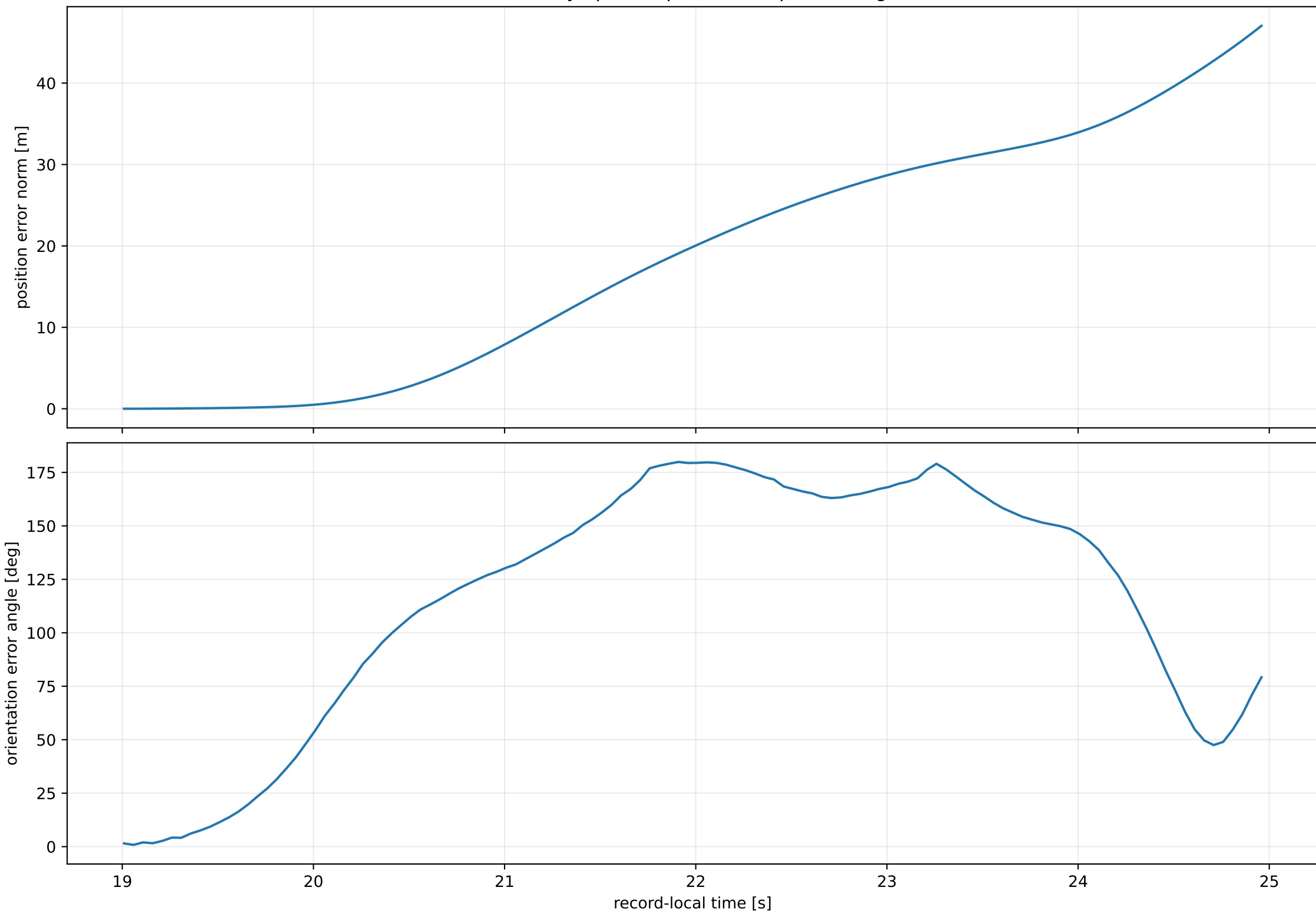
Parameter-only open-loop pose: position



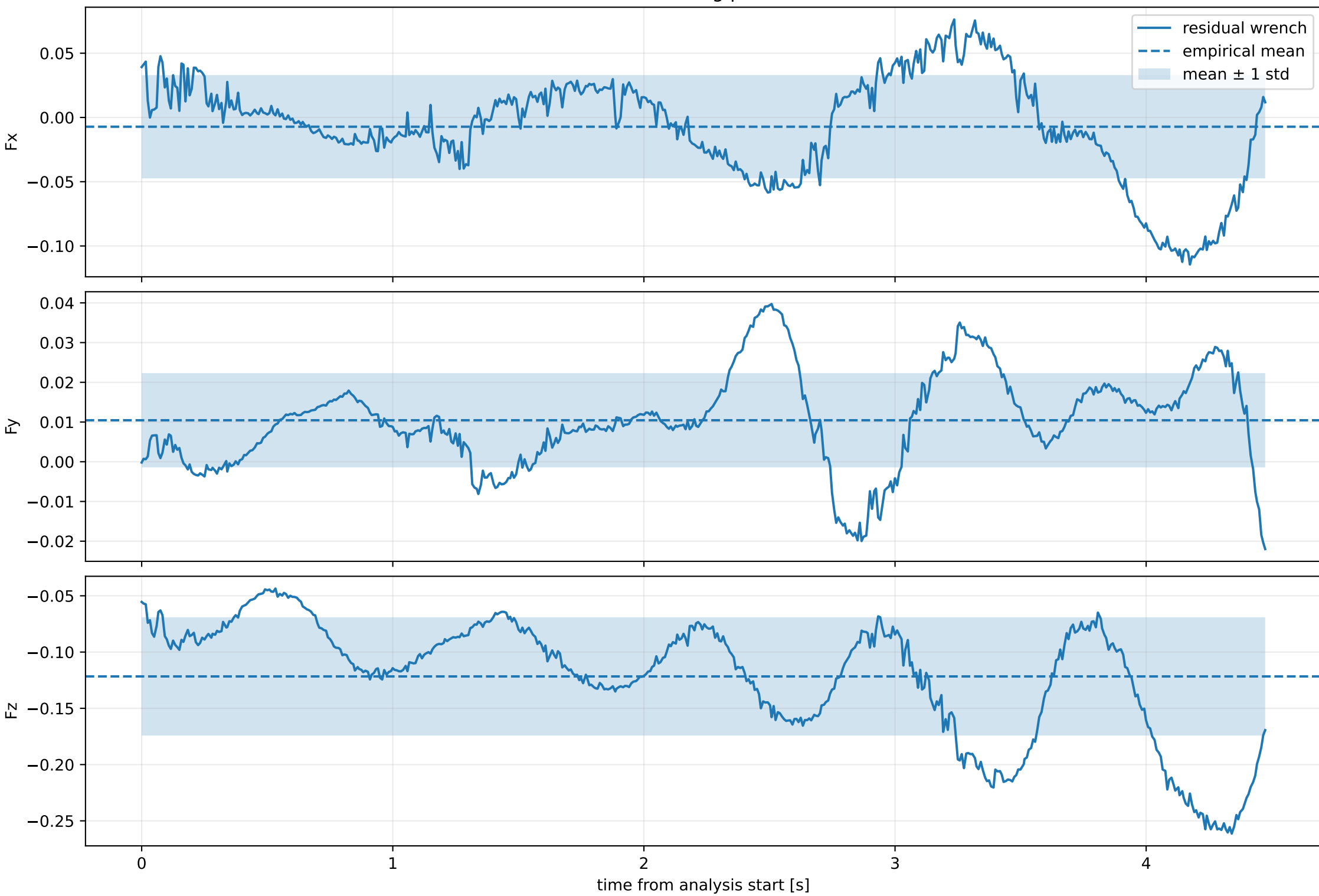
Parameter-only open-loop pose: roll / pitch / yaw



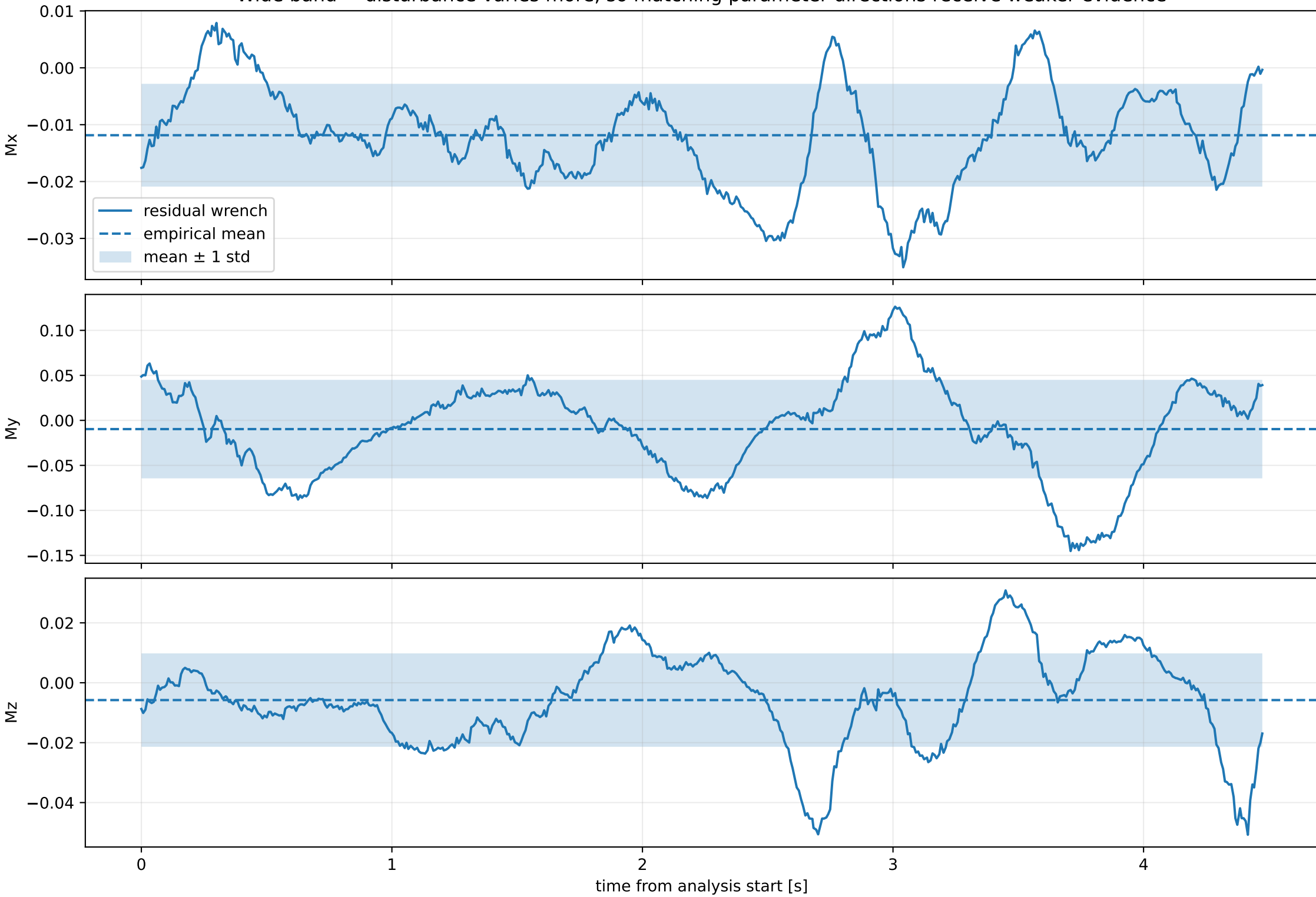
Parameter-only open-loop rollout (separate diagnostic)



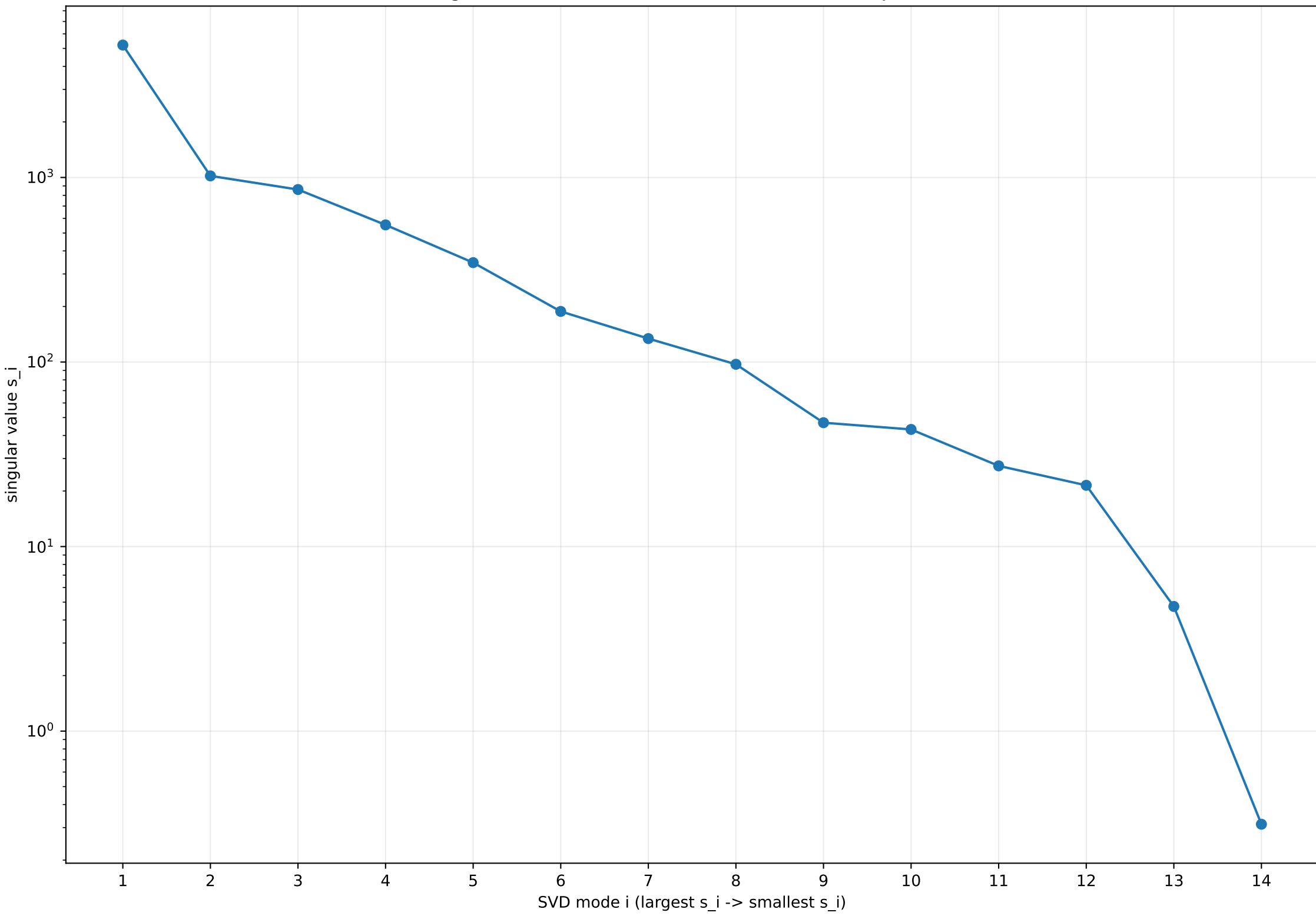
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



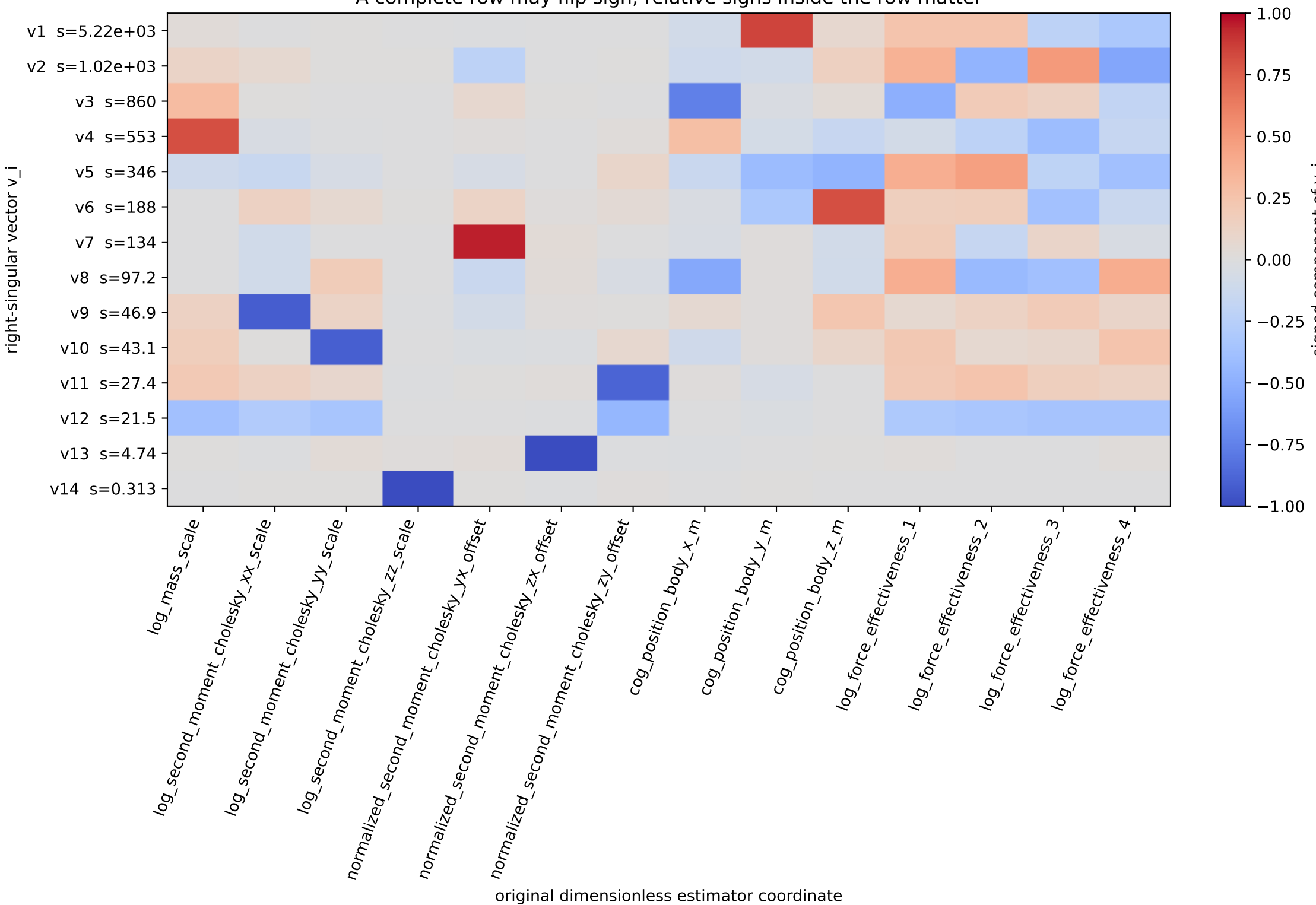
Dimensionless residual torque: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36627839865

inertia J [kg m²]

[0.065011646 -9.8818606e-06 2.314022e-05]
[-9.8818606e-06 0.065242817 6.9237414e-06]
[2.314022e-05 6.9237414e-06 0.12899465]

principal moments [kg m²]

[0.065011216, 0.065243238, 0.12899466]

CoG position in body frame [m]

[-0.0026638041, 0.00086094714, 0.010467908]

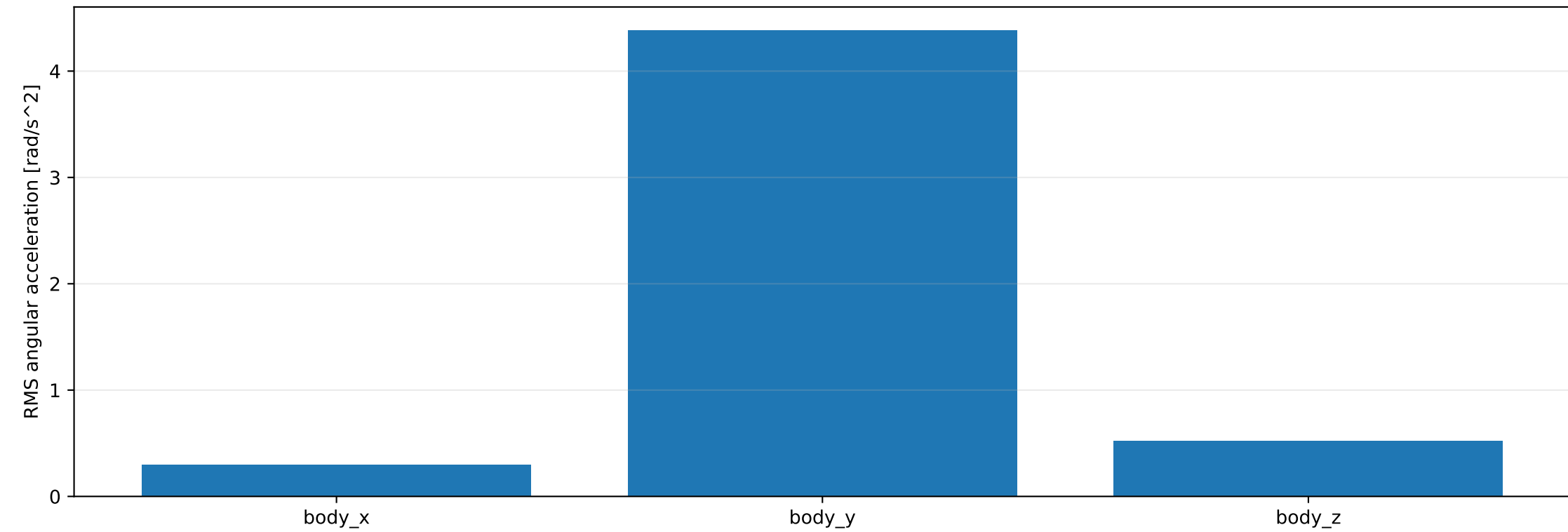
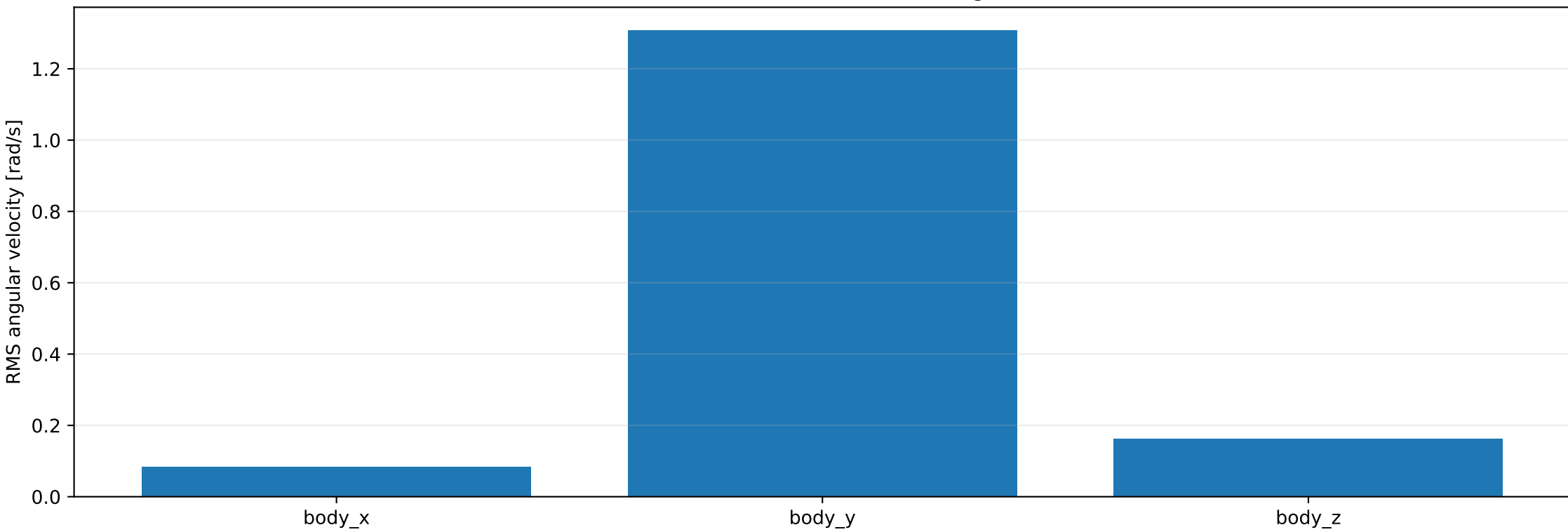
rotor force effectiveness

[0.9623718, 0.97342291, 0.95934074, 0.95180643]

selected delay [s]

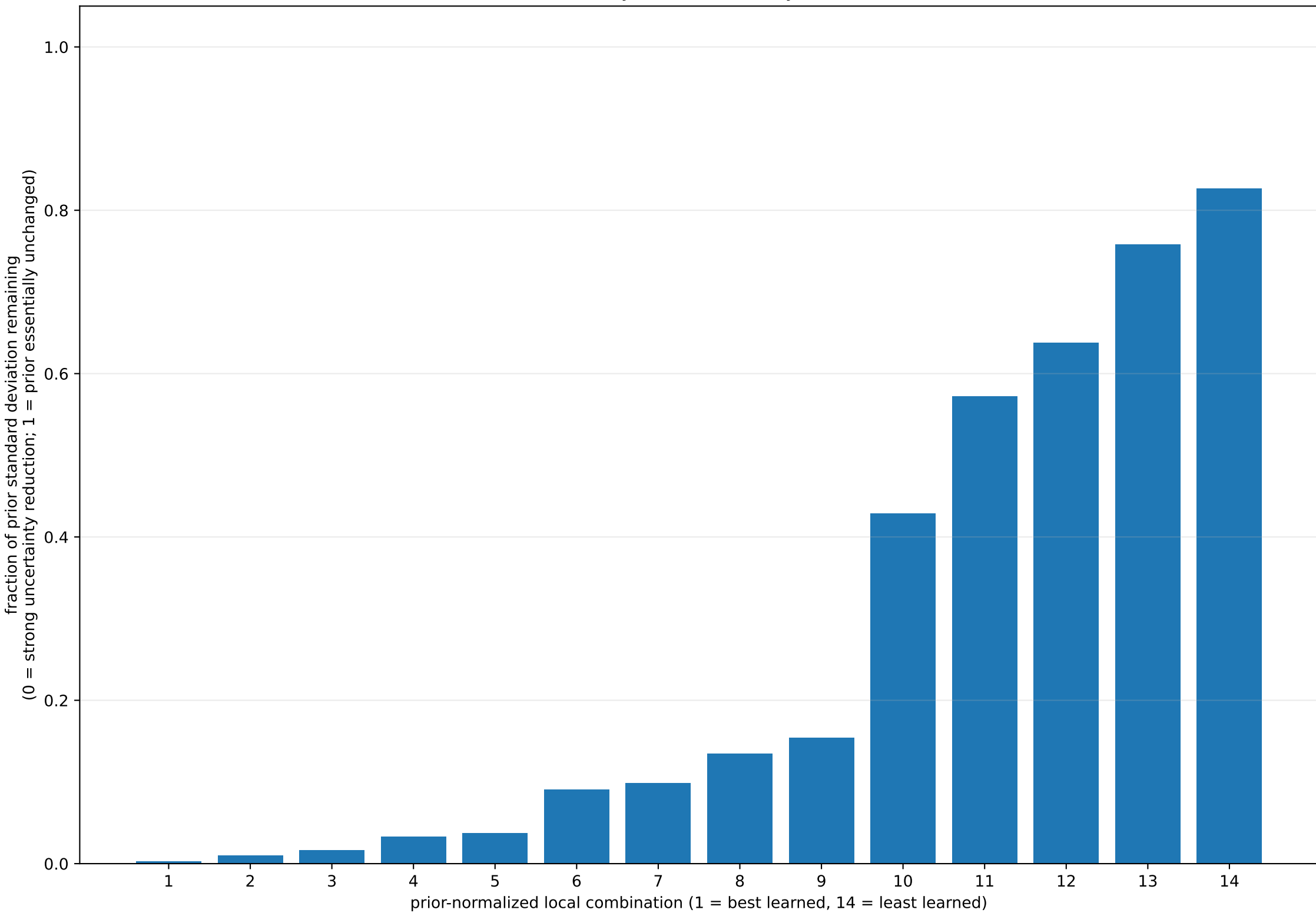
0.190958180968

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

